文献阅读笔记

JAXA 月球着陆任务的月球车部署系统

文献信息 Masataku Sutoh, Takeshi Hoshino, Sachiko Wakabayashi. “Rover deployment system for lunar landing mission.” *Acta Astronautica*, 138 (2017), 454–461. DOI: 10.1016/j.actaastro.2017.06.019。

一句话概括 **论文比较了伸缩式和折叠式两种 3.5 m 月球车下行坡道：伸缩式更轻、更紧凑且可双向展开，折叠式虽需双套、总质量更大，却因机构更简单、故障冗余更高而被认为更适合任务。**

任务约束

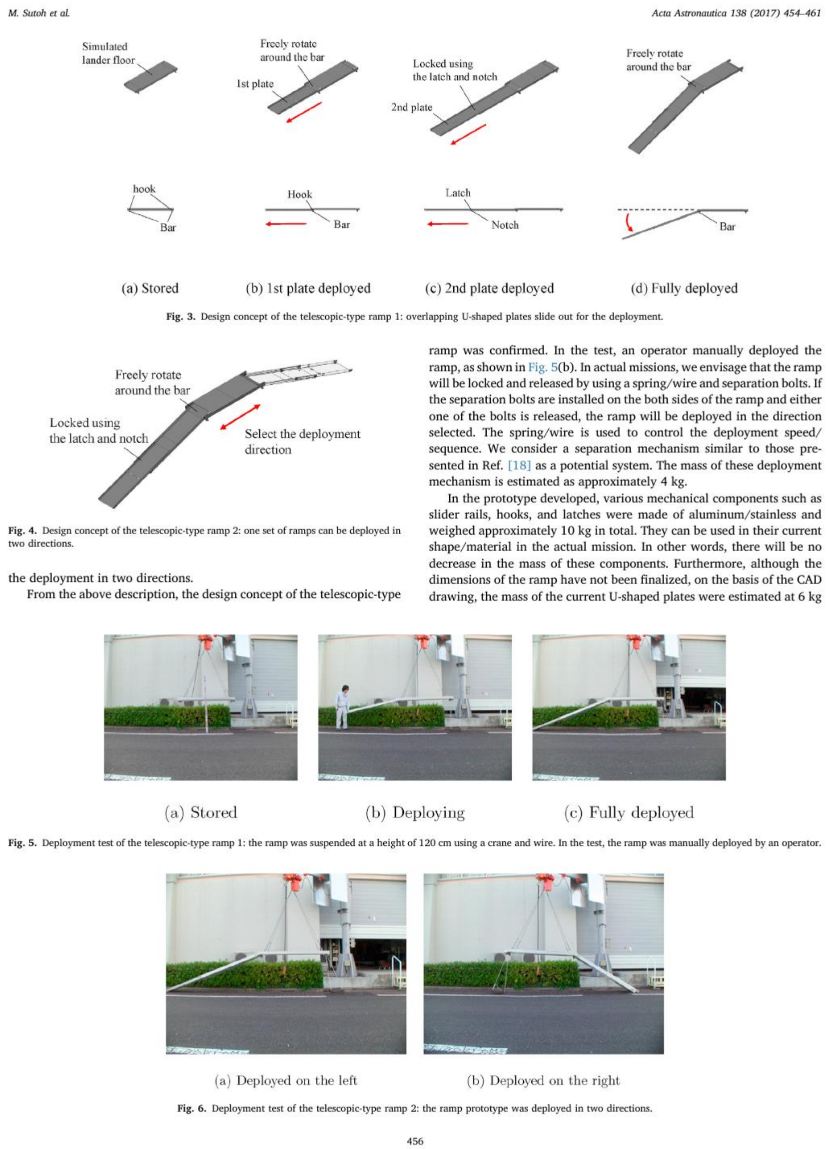
目标是让约 300 kg 月球车从离地 1.2 m 的着陆器平台安全驶下。月球重力约为地球的 1/6，因此坡道设计载荷可用地面约 50 kgf 进行等效验证。为满足月球车通过能力，坡角取 20°，所需长度约 3.5 m；但发射时必须收拢并承受振动。

两种储存长度分别为：伸缩式 1.8 m，折叠式 1.3 m。共同评价指标是质量、安装便利性、可靠性、冗余性、地形适应和月尘影响。

方案 A：伸缩式坡道

多个 U 形板重叠储存，沿侧壁滑轨依次伸出。第一块通过钩与着陆器横杆锁定，第二块通过锁扣/缺口与前一块固定，最后整条坡道在自重下绕横杆倾斜。其突出优点是一套坡道可沿相反方向展开，着陆后可以选择地形更安全的一侧。

原型收拢/展开尺寸为 1.8 × 0.6 m 和 3.5 × 0.6 m，2 mm 铝板原型重约 50 kg。作者估算若换用 CFRP 板并优化结构，飞行版单套可约 20 kg，其中展开机构约 4 kg、滑轨/钩/锁扣等约 10 kg。



(a) Stored (b) 1st plate deployed (c) 2nd plate deployed (d) Fully deployed

Fig. 3. Design concept of the telescopic-type ramp 1: overlapping U-shaped plates slide out for the deployment.

Fig. 4. Design concept of the telescopic-type ramp 2: one set of ramps can be deployed in two directions.

the deployment in two directions.

From the above description, the design concept of the telescopic-type ramp was confirmed. In the test, an operator manually deployed the ramp, as shown in Fig. 5(b). In actual missions, we envisage that the ramp will be locked and released by using a spring/wire and separation bolts. If the separation bolts are installed on the both sides of the ramp and either one of the bolts is released, the ramp will be deployed in the direction selected. The spring/wire is used to control the deployment speed/sequence. We consider a separation mechanism similar to those presented in Ref. [18] as a potential system. The mass of these deployment mechanism is estimated as approximately 4 kg.

In the prototype developed, various mechanical components such as slider rails, hooks, and latches were made of aluminum/stainless and weighed approximately 10 kg in total. They can be used in their current shape/material in the actual mission. In other words, there will be no decrease in the mass of these components. Furthermore, although the dimensions of the ramp have not been finalized, on the basis of the CAD drawing, the mass of the current U-shaped plates were estimated at 6 kg

(a) Stored (b) Deploying (c) Fully deployed

Fig. 5. Deployment test of the telescopic-type ramp 1: the ramp was suspended at a height of 120 cm using a crane and wire. In the test, the ramp was manually deployed by an operator.

(a) Deployed on the left (b) Deployed on the right

Fig. 6. Deployment test of the telescopic-type ramp 2: the ramp prototype was deployed in two directions.

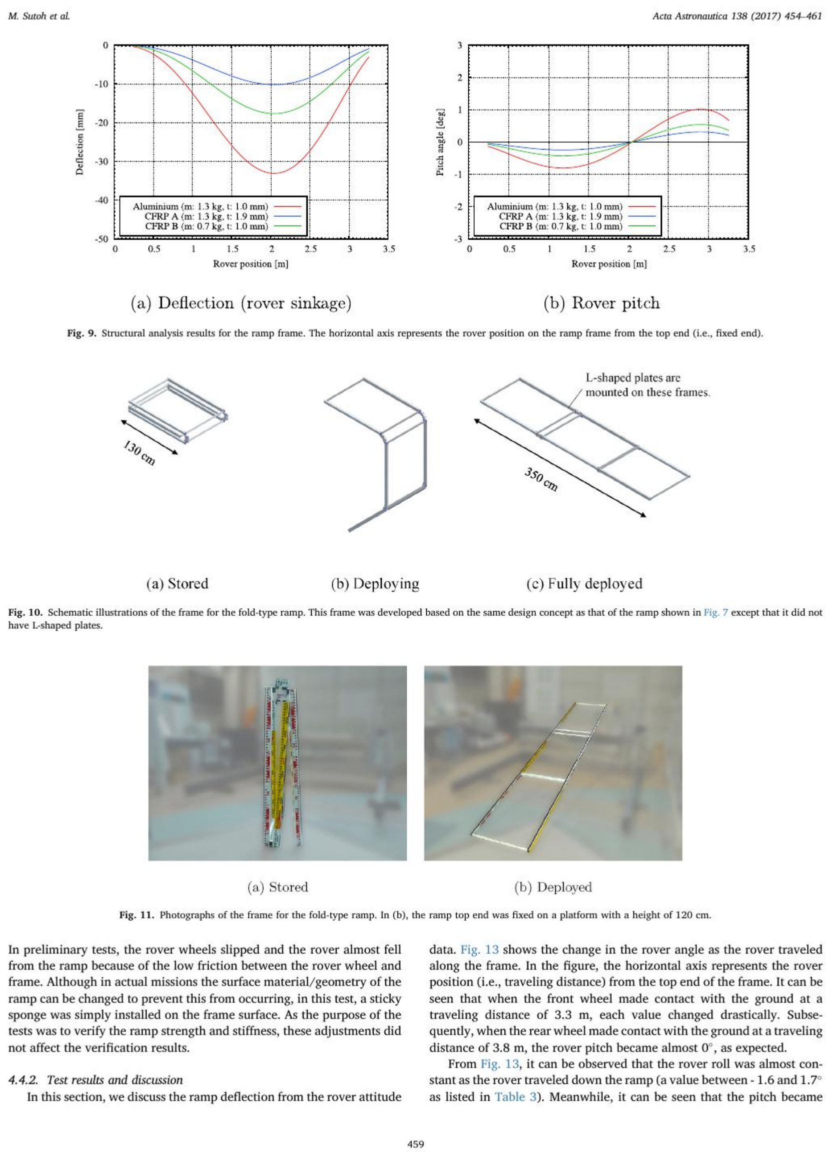
图：原文第 3 页。伸缩结构的最大吸引力是“一套机构、两个可选方向”。

方案 B：折叠式坡道

两次折叠的 L 形路面板由内向外展开。展开完成后，相邻框架端面互相接触并承担竖向载荷。结构分析把单侧框架等效为“一端固支、一端简支”的梁，叠加前后轮集中载荷，计算挠度与月球车俯仰变化。

分析对象为 3.5 m 长、50 × 20 mm 空心矩形截面，比较 1 mm 铝、同质量 CFRP A 和同厚度 CFRP B：

- 铝框最大挠度约 33 mm，理论俯仰变化约 ±1°。
- 同质量 CFRP 可提高刚度；同厚度 CFRP 的质量约为铝的 55%，且俯仰变化更小。
- 铝框架原型质量 5.8 kg；加上 L 形面板和展开机构，单套飞行版估计约 15 kg。



M. Sutoh et al. Acta Astronautica 138 (2017) 454–461

(a) Deflection (rover sinkage) (b) Rover pitch

Fig. 9. Structural analysis results for the ramp frame. The horizontal axis represents the rover position on the ramp frame from the top end (i.e., fixed end).

(a) Stored (b) Deploying (c) Fully deployed

Fig. 10. Schematic illustrations of the frame for the fold-type ramp. This frame was developed based on the same design concept as that of the ramp shown in Fig. 7 except that it did not have L-shaped plates.

(a) Stored (b) Deployed

Fig. 11. Photographs of the frame for the fold-type ramp. In (b), the ramp top end was fixed on a platform with a height of 120 cm.

In preliminary tests, the rover wheels slipped and the rover almost fell from the ramp because of the low friction between the rover wheel and frame. Although in actual missions the surface material/geometry of the ramp can be changed to prevent this from occurring, in this test, a sticky sponge was simply installed on the frame surface. As the purpose of the tests was to verify the ramp strength and stiffness, these adjustments did not affect the verification results.

4.4.2. Test results and discussion

In this section, we discuss the ramp deflection from the rover attitude data. Fig. 13 shows the change in the rover angle as the rover traveled along the frame. In the figure, the horizontal axis represents the rover position (i.e., traveling distance) from the top end of the frame. It can be seen that when the front wheel made contact with the ground at a traveling distance of 3.3 m, each value changed drastically. Subsequently, when the rear wheel made contact with the ground at a traveling distance of 3.8 m, the rover pitch became almost 0°, as expected.

From Fig. 13, it can be observed that the rover roll was almost constant as the rover traveled down the ramp (a value between - 1.6 and 1.7° as listed in Table 3). Meanwhile, it can be seen that the pitch became

459

图：原文第 6 页，包含挠度/俯仰曲线和 3.5 m 折叠框架。

月球车下行试验

地面试验采用 30 kg 四轮车再加 20 kg 配重，等效 300 kg 月球车在月面产生的重力载荷；坡度 20°、速度 1 $cm \cdot s^{-1}$。为防跌落，吊车钢索保持松弛，只提供安全保护；由于轮—框架摩擦不足，试验临时加装黏性海绵。

实测滚转约 –1.6°至 1.7°，偏航约 –1.2°至 1.4°；俯仰为 –25.0°至 –17.9°。最大俯仰比理想梁模型预期更大，作者认为主要来自铰链螺钉松弛和连接处附加变形，而不是框架本体强度不足。

M. Sutoh et al. Acta Astronautica 138 (2017) 454–461

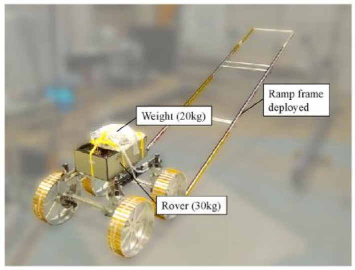


Fig. 12. Photograph of the rover release tests. In the tests, the rover was suspended from a crane using wires to prevent it from falling from the ramp. By manually adjusting the lengths of the wires and keeping them loose, the rover mass was always supported completely by the ramp.

Table 2
Rover specifications. In the rover release tests, the rover mass was set at 50 kg using additional weights.

Item	Value	Unit
Mass	30 (+20)	kg
Wheelbase	50	cm
Track width	65	cm
Wheel diameter	38	cm
Wheel width	16	cm

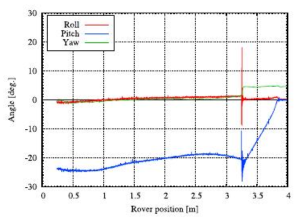


Fig. 13. Rover attitude angle vs. position on frame in the release test. The horizontal axis represents the rover position (i.e., traveling distance) on the ramp frame from its top end.

steeper as the rover traveled from the top end to the midpoint and subsequently became gentler. This trend corresponded with that of the structural analysis shown in Fig. 9(b). The value of the pitch was between - 25.0 and - 17.9°, as listed in Table 3. The yaw was almost constant throughout the test (a value between −1.2 and 1.4°). From the above description, it was confirmed that the frame was not severely deflected.

In the rover release tests using the ramp frame prototype, the rover pitch reached a maximum of 25°. This inclination was larger than that estimated from the structural analysis. In the structural analysis, the frame was assumed to be a long rigid beam without any joints. However, the developed prototype had joints between the frames. Hinges that consisted of joints were fixed to the frame using screws and nuts. When the frame was deformed, the screws may have slackened, resulting in further deformation of the frame. In future tests, the rover pitch could be reduced by improving the joints, e.g., integrated fabrication of the frame and hinges. In addition, the structural analysis shown in Fig. 9 suggests that changing the frame material would also contribute to a decrease in variation of the rover pitch.

In the development of the ramp frame for a 300-kg rover released on the Moon, we believe that no drastic increase in the frame mass from the above prototype will be required. That is, one set of the ramp frames will be developed with a mass of approximately 6 kg. In addition, on the basis of the CAD drawing shown in Fig. 7, the mass of the L-shaped plates was estimated at 5 kg. Based on these estimations, it is suggested that one set of the fold-type ramps could be developed with a mass of approximately 15 kg, including the deployment mechanism and separation bolts. For the fold-type ramp, two sets of the ramps will need to be mounted on the lander. Thus, the total mass of the ramp system is estimated at $30(=15 \times 2)$ kg.

5. Comparison of telescopic- and fold-type ramps

As explained in Section 2, we consider that a mass, ease of mounting, reliability, and redundancy (i.e., fault-tolerance) as important criteria for a rover deployment system. Table 4 provides a comparison of the telescopic- and fold-type ramps in these criteria. From the table, it can be observed that, although the fold-type ramp may be inferior to the telescopic-type in terms of mass and ease of mounting, it is superior in terms of degree of reliability and redundancy. In this section, the reasons of the rating in the table is explained in detail.

For the telescopic-type ramp, while its prototype weighted approximately 50 kg, the mass of the final system was estimated at 20 kg. This mass was estimated on the assumption that the dimension and material can change for the final system. On the other hand, for the fold-type ramp, while its ramp frame prototype weighted approximately 6 kg, the mass of the one set of final system was estimated at 15 kg. For two sets of the ramp, the total mass is 30 kg. The mass was increased in the final system because only the ramp flames were developed as the prototype and various additional components are essential in the final system. Based on the expected mass of the final systems, the telescopic-type ramp will be lighter than the fold-type ramp. For this reason, a mass was rated as "good" for the telescopic-type ramp in Table 4.

The telescopic-type ramp can be stored under the lander floor and one set of the ramps can be deployed in two directions. Meanwhile, two sets of the fold-type ramps need to be mounted on the front and rear ends of the lander. Thus, the total volume of the fold-type ramp is larger than that of the telescopic-type ramp. Furthermore, considering the storage configuration, it seems that the telescopic-type ramp can be more compact and have less impact on the lander total design than the fold-type ramp. For this reason, an ease of mounting was rated as "good" for the telescopic-type ramp in Table 4.

Note that regolith affects mechanisms of each ramp on the lunar surface. For the telescopic-type ramp, the regolith could become adhered

Table 3
Average, minimum, and maximum values of the rover attitude angles in the release test. Each value was obtained from the data before the front wheel of the rover made contact with the ground at the rover position of 3.3 m shown in Fig. 13.

	Roll [°]	Pitch [°]	Yaw [°]
Average	0.4	−21.4	0.03
Minimum	−1.6	−25.0	−1.2
Maximum	1.7	−17.9	1.4

460

图：原文第 7 页。试验揭示了"理想连续梁"与"带铰链真实结构"的差异。

为什么最后偏向折叠式

维度	伸缩式	折叠式
估计总质量	约 20 kg/单套	约 15 kg/套，但需两套，共约 30 kg
收纳/安装	更紧凑，可藏于底板下	占空间更大，需在两侧安装

维度	伸缩式	折叠式
地形方向选择	单套可双向部署	每套单向，靠两套冗余
月尘敏感性	滑轨可能被月壤卡滞	铰链结构相对直接
故障后果	单套失败则无法下车	一侧失败仍可用另一侧

航天任务把可靠性和冗余性的优先级放在质量与安装便利性之上，所以作者把折叠式视为更有前景的方案。这是典型的“不是单项最优，而是任务风险最优”。

局限与后续工作

- 两种坡道均为手动展开原型，真实分离螺栓、弹簧/钢索、速度控制和月面锁定尚未形成完整飞行机构。
- 地面等效仅匹配静态重力载荷，不能完全复制月面接触、惯性、真空润滑、热循环和月尘。
- 折叠坡道的临时海绵说明路面摩擦设计仍未闭环；真实踏面材料和轮齿啮合需要单独验证。
- 最值得继续的方向是铰链一体化、CFRP 优化、月尘容错和部署动力学仿真，并以失效模式/Fault Tree 驱动双侧冗余设计。

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